A PROJECT REPORT ON

Soil Quality Classification Using Decision Trees and Random Forest

A project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

A.Y.: 2026-27

Submitted by:

N. Divya Sree – 2520090031

D. Sri Varshini – 2520080013

Under the Esteemed guidance of

I.V. Sai Lakshmi Haritha

Assistant Professor, CS&IT

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Soil Quality Classification Using Decision Trees and Random Forest

1. ABSTRACT

Soil quality is one of the most important factors in agriculture because it affects crop growth and production. Healthy soil helps farmers grow better crops, while poor-quality soil can reduce crop yield. Usually, checking soil quality requires laboratory testing, which can take time and cost money. Machine Learning provides a faster and smarter way to analyze soil and predict its quality.

The main aim of this project, "Soil Quality Classification Using Decision Trees and Random Forest," is to classify soil quality using machine learning techniques. The system uses soil properties such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, moisture, organic carbon, and other important features to predict the quality of soil. This project follows the complete Machine Learning process, starting from collecting the dataset, cleaning the data, preparing the features, training different machine learning models, evaluating their performance, and finally deploying the best model for prediction. Logistic Regression is used as a basic model, while Decision Tree and Random Forest are used as the main classification algorithms. K-Means Clustering and Principal Component Analysis (PCA) are also used to understand and visualize the dataset.

The performance of each model is compared using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The best-performing model is then used to predict soil quality through a simple user interface. This project helps farmers, students, and researchers make better decisions about soil management and crop selection. It also demonstrates the complete machine learning workflow, from data preparation to model deployment, using simple and effective machine learning techniques.

2. INTRODUCTION

Agriculture is an important part of our daily life because it provides food and supports the economy. The quality of soil plays a major role in growing healthy crops. Soil contains nutrients that plants need for proper growth. If the soil quality is good, crops grow better and produce higher yields. Traditionally, soil quality is checked through laboratory testing. Although these tests are accurate, they require time, money, and expert knowledge. Machine Learning can make this process faster by analyzing soil data and predicting soil quality automatically.

This project focuses on developing a machine learning system that classifies soil quality using Decision Tree and Random Forest algorithms. The system uses different soil properties as input and predicts the quality of the soil. Logistic Regression is also used to compare the performance of different machine learning models. K-Means Clustering and PCA help in understanding the data and finding useful patterns. The project includes all the important steps of the machine learning process, such as data collection, data preprocessing, feature engineering, model training, model evaluation, and deployment. The models are tested using different evaluation methods to find the most accurate one. The main goal of this project is to build a simple, accurate, and reliable system that can classify soil quality. This project also helps students understand how machine learning techniques can be applied to solve real-world agricultural problems.

3. SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10 or Windows 11 / Ubuntu Linux (Optional)
- **Programming Language:** Python 3
- **Development Tools:** Jupyter Notebook, Google Colab, Visual Studio Code (VS Code)
- **Python Libraries:** Pandas (Data Management), NumPy (Math Calculations), Scikit-learn (ML Models), Matplotlib & Seaborn (Visualization), Joblib (Model Saving)
- **Machine Learning Algorithms:** Logistic Regression, Decision Tree, Random Forest, K-Means Clustering, Principal Component Analysis (PCA)
- **Deployment Tool:** Flask or FastAPI
- **Dataset:** Soil Quality Dataset containing N, P, K, pH, moisture, organic carbon, and related features.

Signature of the Guide **Course Instructor**